

A Discrete-Component Second-Order CT Sigma-Delta ADC Design

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1 Overview

In the sigma-delta ($\Delta\Sigma$) ADC topology, a quantizer produces a stream of coarse 1-bit decisions, each of which is fed back through a 1-bit DAC into the loop’s cascaded integrators, shaping the input to subsequent decisions. Because resolution is built up from many such decisions rather than a single high-precision comparison, a $\Delta\Sigma$ ADC trades conversion latency for resolution relative to a SAR ADC, which resolves a full-precision word in a single conversion cycle. This report documents a second-order continuous-time sigma-delta ADC, designed and tested in LTspice before discrete breadboard implementation using physical components. The target specifications for the design are listed in Table 1.

Table 1: Design Specifications

Parameter	Value
Signal Bandwidth	200 Hz
Oversampling Ratio (OSR)	128
Sample Rate	51.2 kHz

2 Architecture

3 Bring-Up Methodology

3.1 Root Cause: DAC Sign Error

3.2 Debugging Highlight: Instrumentation Artefacts

4 Results

5 Limitations

6 Next Steps

References

A Source Code and Reproducibility